Supplementary Material

WaveCatBoost for Air-Quality Forecasting

Applied Time Series Analysis — Course Project, Group 8

Kumar Utkarsh P. Sai Deva Charan M. Charan Sree Teja D. Lalit Santosh

This one-page note points to the code, interactive demo, and written report that accompany our presentation. All linked material implements or explains Borah et al. (IEEE Sensors Letters, 2025) and our course extensions.

Code repository

- GitHub (project source, experiments, report, slides): https://github.com/charan-435/Forecasting-Air-Quality-Data
- Suggested branch for the current submission: `kaisen`
- Paper author code (reference implementation): https://github.com/jborah2/WaveCatBoost-Time-Series-Model

Interactive visualisation / demo (encouraged)

- **WaveCatBoost Lab** (datasets, EDA, MODWT, CatBoost, results, conformal, extensions): https://charanponaganti.github.io/wavecatboost-learn/
  Local run: `cd lab && npm install && npm run dev`
- Lightweight recorded-result web demo: `presentation/webapp/` (open via a local static server; scores from `results/paper/scores.csv`)

How to implement (paper method + ours)

- Interactive report: `analysis.Rmd` (open in RStudio → Run Document)
- Experiment entry points: `scripts/experiments/` (`run_experiments.R`, filter / Fourier / EMD / Shillong runners)
- Core pipeline: `R/core/` (MODWT, models, metrics, conformal)
- Setup and reproduce commands: repository `README.md`

Presentation and recorded results

- Course deck: `presentation/main.pdf`
- Recorded recursive bake-off scores: `results/paper/scores.csv`
- Paper-protocol 7-day panels / residual PI: `results/paper/paper_protocol.rds`

Note. Headline scores are measured recursive multi-step forecasts from our pipeline, not the letter's rolling 1-step Table 1. Theoretical proofs are not part of this applied project.